

Project/Thesis No.:

CSE 4000: Project/Thesis

**DEVELOPMENT OF A FRAMEWORK FOR AUTOMATED REPORT
GENERATION FROM X-RAY IMAGING USING
RETRIEVAL-AUGMENTED GENERATION AND VISUAL QUESTION
ANSWERING WITH VLM**

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August, 2025

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A thesis submitted in partial fulfillment of the requirements for the degree of
“Bachelor of Science in Computer Science & Engineering”

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Acknowledgement

I want to start by thanking Almighty for always being there for me and for bestowing upon me all of his blessings. I would like to take this opportunity to thank my supervisor, Md. Badiuzzaman Shuvo, Lecturer in the Department of Computer Science and Engineering (CSE), Khulna University of Engineering & Technology (KUET), for his unwavering support, constant direction, and vigilant oversight throughout this study. I am extremely grateful to him for giving me his valuable time whenever I need it and constantly continuing to encourage me in my efforts.

Lastly, I want to express my gratitude to my parents, relatives, and friends for their understanding, encouragement, and support throughout this time.

Author

Abstract

The objective of this research is to explore how large language models (LLMs) can be applied to enhance chest X-ray (CXR) interpretation. The proposed methodology involves integrating multi-modal architectures with multiple task-specific datasets to support diverse functions such as medical report generation (MRG), visual grounding, and visual question answering (VQA). By unifying these tasks in a conversational framework, the system is expected to provide more accurate and clinically relevant insights than existing approaches. The research begins by analyzing the limitations of current AI-based medical imaging methods, particularly the underutilization of LLM multi-tasking capabilities and the lack of clinical precision. This understanding guides the development of a multi-modal framework that addresses these gaps. Ethical considerations related to patient privacy, fairness, and clinical safety are embedded into the design to ensure responsible and trustworthy outcomes. The project timeline includes phased implementation, starting with controlled experiments on static CXR datasets and progressing toward adaptation for real-world clinical scenarios. Expected outcomes include measurable improvements in diagnostic accuracy, efficiency, and the reduction of reporting time. Through comprehensive experimentation and benchmarking, the study aims to quantify the performance gains achieved by multi-modal LLM integration. The findings have implications beyond radiology, extending to broader healthcare applications where efficiency, reliability, and fairness in AI-driven decision-making are critical.

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CHAPTER I

Introduction

1.1 Background

Chest X-rays are one of the most common diagnostic tools for detecting thoracic diseases, yet interpretation is often difficult due to low contrast, overlapping structures, and subtle abnormalities. These challenges can lead to variability in radiologist reports and hinder automated analysis. Recent progress in deep learning and vision–language models (VLMs) has enabled automatic report generation, but issues such as hallucinations, limited contextual reasoning, and high computational cost remain. Retrieval-Augmented Generation (RAG) offers a potential solution by combining visual features with external medical knowledge to generate reports that are more accurate, reliable, and clinically meaningful.

1.2 Problem Statement

In medical imaging, automatic generation of chest X-ray (CXR) reports is vital for supporting clinical diagnosis. A key challenge is building systems that annotate findings accurately while minimizing false positives, since errors may cause unnecessary patient anxiety. Another challenge is real-time reporting, where automated systems must provide immediate insights during examinations to aid timely decisions.

Equally important is creating user-friendly interfaces that allow clinicians to interact effectively with automated tools, ensuring seamless integration into medical workflows. Rising demand for fast, reliable diagnostic support highlights the limits of current methods, which often struggle with accuracy and efficiency. To address these issues, advanced machine learning and natural language processing are being applied to improve both precision and interpretability in automated CXR reporting. Moreover, incorporating retrieval-augmented techniques enables models to reference prior cases, enhancing contextual understanding and clinical relevance.

1.3 Objectives

The objectives of automatic medical report generation encompass a blend of technical innovations and practical applications, which include enhancing diagnostic accuracy, reducing radiologist workload, and improving accessibility of medical services. Furthermore, these objectives aim to ensure consistency in reporting and support clinical decision-making across diverse healthcare settings, which include:

1. **Image understanding:** To develop a model that can understand the visual content of an image.
2. **Question understanding:** To develop a model that can understand the meaning of natural language questions.
3. **Data Collection:** To develop a comprehensive set of information to ensure accurate and meaningful interpretations.
4. **Reasoning:** To develop a model that can reason about the visual content of an image and the information contained in the question to generate accurate answers.
5. **Multi-task Learning:** To generate a model to simultaneously learn multiple related tasks, such as disease classification, localization etc.
6. **Performance Evaluation:** To establish evaluation metrics to assess the automatic report generation method's performance in terms of accuracy and speed. Furthermore, continuously refining the model based on evaluation results.

In essence, the objectives of automated medical report generation revolve around the technical challenges of data extraction, natural language processing, and report accuracy, while also emphasizing its practical usability to ensure a comprehensive and effective system.

1.4 Scope

The scope of creating a automatic medical report generation model is to build a system that can process both visual content and textual input to generate accurate answers. This involves

integrating computer vision and natural language processing techniques. The required tools and components for creating a report generation model include:

1. **Image Feature Extraction:** The report will cover methods for extracting key features from chest X-ray images using deep learning models such as Vision Transformers (ViT-based encoders).
2. **Disease Detection and Classification:** It will address the development of automated techniques for detecting and classifying diseases (e.g., Atelectasis, Cardiomegaly, Effusion) with high diagnostic accuracy.
3. **Report Generation:** The focus will include Natural Language Processing (NLP) approaches for generating structured and clinically meaningful reports based on extracted image features and model predictions.
4. **Data Preprocessing and Augmentation:** The scope extends to preprocessing tasks such as image normalization, resizing, and augmentation, ensuring data quality and robustness for model training.
5. **Cross-Modal Integration:** The study will explore methods to combine visual features with clinical metadata (e.g., patient history and demographics) for enhancing contextual relevance and interpretability.
6. **Limitations and Future Directions:** Finally, the report will highlight challenges faced in automated medical report generation and propose directions for future improvements in model accuracy, usability, and clinical deployment.

Through the exploration of these areas, this thesis aims to contribute to the advancement of efficient parallel automatic report generation techniques, particularly in the context of medical imaging using RAG.

1.5 Unfamiliarity of the Problem

The unfamiliarity of the problem in automated chest X-ray report generation arises from several critical challenges. Firstly, extracting effective information from visual data for Large Language Models (LLMs) to improve results is a complex and pressing issue. LLMs

require high-quality, diverse datasets to generalize effectively, but the rarity of certain diseases and abnormal conditions significantly limits model performance. Additionally, bridging visual patterns in medical images with informative human linguistic descriptions poses difficulties due to the inherent complexity of medical images and the subtle nature of some abnormalities. Furthermore, the process of report writing itself presents challenges. For less experienced physicians, writing detailed and accurate reports can lead to errors, while for seasoned professionals, the task is often time-consuming and tedious. Identifying abnormal regions in chest X-rays, particularly when the variations are subtle, adds another layer of complexity. These challenges highlight the unfamiliar and intricate nature of the problem, requiring sophisticated solutions that blend visual data analysis with natural language processing and medical expertise.

1.6 Project Planning

This project plan outlines the process for automated chest X-ray report generation. The overall project timeline, up to the thesis defense, is illustrated in Figure 1.1.

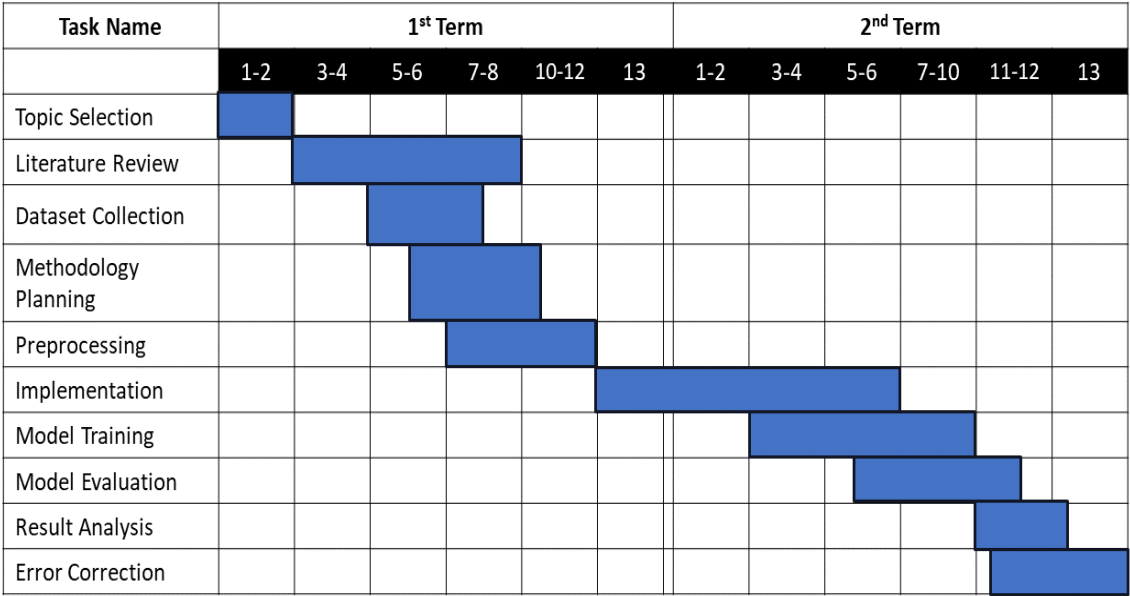


Figure 1.1: Project timeline for automated chest X-ray report generation

1.7 Application of the work

In the field of automatic report generation for chest X-ray datasets, this work focuses on applying machine learning and natural language processing (NLP) techniques to generate descriptive and diagnostic reports directly from medical images. The integration of deep learning for visual feature extraction and language models for report generation opens up several practical applications, including the following.

1. **Automated Diagnosis and Report Generation:** AI-driven systems can assist radiologists by reducing workload and providing reliable second opinions, contributing to more consistent and efficient interpretations.
2. **Real-Time Clinical Support:** In emergency care or high-volume radiology settings, real-time automated reporting can help address radiologist shortages and accelerate diagnostic workflows.
3. **Reducing Diagnostic Errors:** Acting as a decision support system, automated reporting can improve diagnostic accuracy by cross-verifying initial radiologist reports, thereby minimizing human error.
4. **Radiology Education and Training:** Such systems can be valuable in medical education, enabling students to compare their interpretations with AI-generated reports and better understand diagnostic patterns.
5. **Healthcare Communication:** Automated reports can enhance communication with patients by providing clear and consistent descriptions of imaging findings.

1.8 Organization of the report

The remainder of the paper is organized as follows. **Chapter II** reviews related work in the domain. **Chapter III** describes the proposed methods in detail. **Chapter IV** presents experimental studies, including the setup, dataset preparation, implementation, and results, along with discussions on achieved objectives, ethical considerations, socio-economic impact, sustainability, and financial analysis. **Chapter V** examines social, health, and other related issues. **Chapter VI** discusses complex engineering activities. Finally, **Chapter VII** concludes the paper and outlines future research directions.

CHAPTER II

Literature Review

2.1 Literature Review

Automatic Report Generation has been a major research area for the past few years and is undoubtedly one of the most popular visual reasoning problems [1]. Such tasks require a deep understanding of both the visual content and associated textual information, making them essential for progressing in areas like medical image analysis and automated diagnosis [2]. Current methodologies often leverage powerful deep learning architectures to extract visual features and generate coherent, contextually accurate reports [3].

2.1.1 Automatic Report Generation Architecture: Vision Backbone + LLMs

A key challenge in medical report generation is improving LLM outputs while reducing the high computational cost of visual transformers. To address this, Jing et al. [4] propose a context-guided framework using Mamba as the vision backbone, offering linear complexity with transformer-level performance. Vision tokens, context, and prompts are then fed to the LLM for high-quality report generation. Experiments on IU-Xray, MIMIC-CXR, and CheXpert Plus validate the model’s effectiveness. The figure 2.1 shows the overall architecture for generating the report. This approach demonstrates a promising balance between computational efficiency and diagnostic accuracy, making it feasible for practical deployment. Furthermore, the framework highlights the potential of combining efficient vision backbones with advanced language models to set new benchmarks in medical report generation.

2.1.2 Multi-Tasking of LLMs: CNN + LLMs + VQA

In chest X-ray (CXR) analysis, previous studies have utilized large language models (LLMs), but they often face limitations, either by underutilizing the multi-tasking

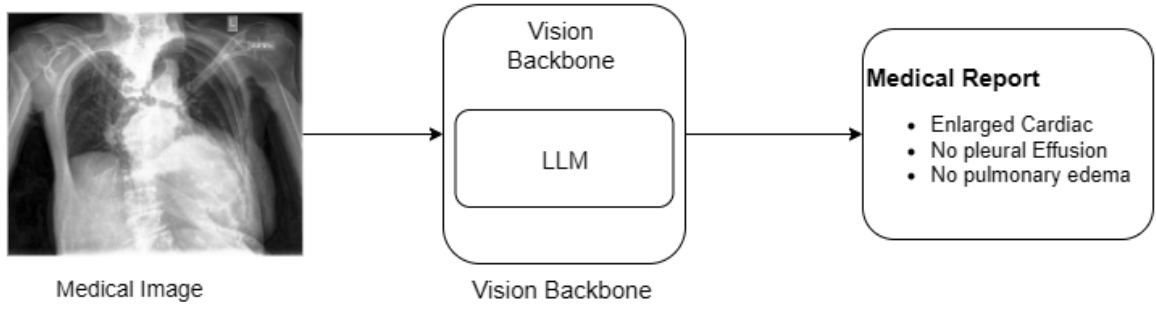


Figure 2.1: Report Generation Using Vision Backbone and LLMs

capabilities of LLMs or lacking clinical accuracy. Park et al. [5] presents M4CXR, a multi-modal LLM specifically designed to enhance CXR interpretation. The model is trained on a visual instruction-following dataset that integrates various task-specific datasets in a conversational format.

2.1.3 Minimizing Cross Entropy Loss: GAN

Previous medical report generation models have primarily focused on minimizing cross-entropy loss or optimizing common image-captioning metrics, such as CIDEr, often overlooking diagnostic accuracy, which should be the primary consideration in this field. These approaches tend to prioritize linguistic similarity to reference reports rather than ensuring that the generated reports reflect the underlying medical conditions accurately. Inspired by generative adversarial networks, Shin et al. [6] propose an adversarial reinforcement learning approach for generating reports from chest X-ray images that considers both diagnostic accuracy and language fluency. Specifically, an accuracy discriminator (AD) and a fluency discriminator (FD) are developed to evaluate the generated reports based on these two criteria. The AD guides the model to generate clinically relevant findings, while the FD ensures that the report remains coherent, natural, and readable. This dual-discriminator strategy not only strengthens the clinical relevance of the generated reports but also enhances the reliability and trustworthiness of automated report generation systems. Furthermore, this approach highlights the importance of integrating domain-specific evaluation metrics into the training process to better align model outputs with real-world diagnostic needs. As a result, such methods represent a significant step towards deploying AI-assisted tools in clinical practice, where both accuracy and interpretability are critical.

2.1.4 Bridging Visual Patterns with informative descriptions: CNN + Encoder + Decoder

Generating long and coherent reports to describe medical images presents challenges in bridging visual patterns with informative human linguistic descriptions. Fisman et al. [7] propose a novel Hybrid Retrieval-Generation Reinforced Agent (HRGR-Agent) that reconciles traditional retrieval-based approaches, which incorporate human prior knowledge, with modern learning-based methods to achieve structured, robust, and diverse report generation.

2.1.5 Report Generation: CNN + LSTM

Medical imaging report writing is error-prone for novices and tedious for experts. To address this, Yuan and Liang et al. [8] explore automatic report generation, a task complicated by diverse information needs, difficulty in detecting abnormalities, and lengthy multi-sentence

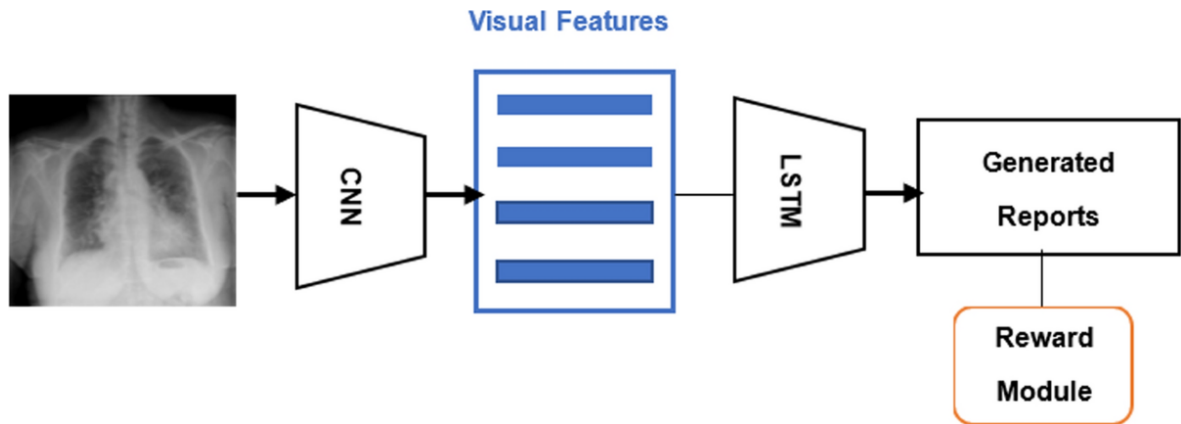


Figure 2.2: Demonstrate Report Generation using CNN + LSTM

outputs. Their approach introduces (1) a multi-task learning framework for tag prediction and paragraph generation, (2) a co-attention mechanism to localize abnormalities, and (3) a hierarchical LSTM for producing long reports. Figure 2.2 shows the overall architecture for demonstrating report generation using CNN + LSTM.

2.2 Discussion of research gap solution

The table 2.1 provides a concise overview of recent studies in the field of X-ray report generation. While medical report generation algorithms have been explored, there appears

to be limited research on generating reports that incorporate detailed knowledge of various diseases. Most existing methods primarily emphasize generic image-to-text translation without integrating domain-specific clinical knowledge. This gap highlights the need for frameworks that not only describe visual findings but also contextualize them with disease-specific information, thereby improving diagnostic accuracy and clinical usability.

Table 2.1: Recent Studies in the Domain

Reference	Method	Findings	Limitations
arXiv: 2408.09743 (2024) [9]	Retrieving Context Samples for Large Language Model based X-ray Medical Report Generation	Extract more effective information for the LLMs.	The knowledge about the disease is ignored
arXiv: 1711.08195 (2017) [10]	On the Automatic Generation of Medical Imaging Reports	Generate long texts that contain multiple sentences or even paragraphs.	Long descriptive paragraphs.
arXiv: 2408.16213 (2024) [11]	Exploring Multi-task Potentials of Multi-modal Large Language Models for Chest X-ray Interpretation	Improving clinical accuracy.	Low NLG metrics and the presence of hallucinations.
IEEE Access et al. (2021) [12]	Report Generation for Chest X-Ray Images via Adversarial Reinforcement Learning	Enhancing diagnostic accuracy and language fluency.	The model proposed fully learnable ARL architecture
Advances in neural information processing systems (2018) [13]	Hybrid Retrieval-Generation Reinforced Agent for Medical Image Report Generation	Improving bridging visual patterns with informative human linguistic descriptions.	Majority reports based on normal case.

CHAPTER III

Methodology

3.1 Definitions

The main objective is to develop a medical report generation system that combines the most effective features from various approaches proposed by leading researchers. This system leverages insights from previous studies while incorporating detailed knowledge of a broad range of diseases. By integrating these advancements, the approach aims to generate more accurate and comprehensive medical reports, addressing both common and less frequently studied conditions.

We propose a two-stage framework that (1) fine-tunes a vision–language model (VLM) on chest X-ray $\rightarrow$ text supervision using PEFT (LoRA/QLoRA), and (2) wraps the fine-tuned model in a Retrieval-Augmented Generation (RAG) pipeline backed by FAISS with MiniLM embeddings to ground outputs to prior similar cases and clinical guidance.

3.2 Fine-Tuning the VLM with PEFT (LoRA / QLoRA)

Large Vision–Language Models (VLMs) such as BLIP or BLIP-2 contain hundreds of millions of parameters, enabling them to capture complex relationships between visual and textual modalities. However, full fine-tuning of all parameters is computationally expensive and often infeasible in resource-constrained environments, such as standard workstations or cloud-based GPUs with limited memory. Parameter-Efficient Fine-Tuning (PEFT) addresses this challenge by updating only a small subset of parameters while keeping the majority of the backbone model frozen. This not only reduces the computational cost but also lowers the risk of overfitting, especially when training on domain-specific datasets with limited examples. Two widely adopted PEFT methods are Low-Rank Adaptation (LoRA) and Quantized LoRA (QLoRA). Table 3.1 illustrates the overall workflow for fine-tuning the Vision–Language Model (VLM) on the CheXpert dataset, highlighting the integration

of PEFT methods for efficient and scalable training.

3.2.1 LoRA: Low-Rank Adaptation

In transformer models, the largest parameter matrices are often the weight matrices of the self-attention mechanism (e.g., query, key, value, and output projections). Instead of fully updating these matrices during fine-tuning, LoRA decomposes their weight updates into two much smaller trainable matrices.

Let $W_0 \in \mathbb{R}^{d \times k}$ be a frozen pre-trained weight matrix. During LoRA fine-tuning, its update is parameterized as:

$$W = W_0 + \Delta W, \quad \Delta W = AB \quad (3.1)$$

where:

$$A \in \mathbb{R}^{d \times r}, \quad B \in \mathbb{R}^{r \times k}, \quad r \ll \min(d, k)$$

Thus, instead of learning $d \times k$ parameters, we only learn $r(d + k)$ parameters, which is a significant reduction.

For example, if $d = k = 1024$ and $r = 16$:

- Full fine-tuning: 1,048,576 parameters.
- LoRA: 32,768 parameters.

This is approximately 32× more efficient.

3.2.2 QLoRA: Quantized Low-Rank Adaptation

QLoRA further improves efficiency by quantizing the base model into 4-bit NormalFloat (NF4) format, reducing GPU memory consumption without harming accuracy. During training:

- The frozen backbone weights are stored in 4-bit precision.
- The LoRA adapters are trained in higher precision (e.g., bfloat16).
- Double quantization and paged optimizers are used to reduce memory spikes.

This allows fine-tuning of models with billions of parameters on GPUs as small as 24–48 GB. Figure 3.1 illustrates the fine-tuning methods along with their corresponding memory requirements.

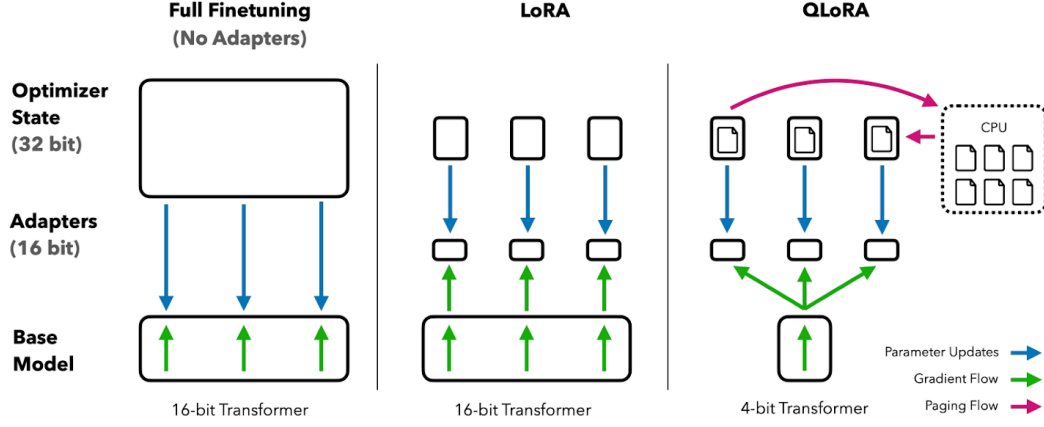


Figure 3.1: Fine tuning Methods and their memory requirements

3.2.3 Training Objective

For image-to-text training (e.g., generating a radiology report from an X-ray), the objective is causal language modeling:

$$\mathcal{L}_{CE} = - \sum_{t=1}^T \log P_{\theta}(y_t | y_{<t}, v) \quad (3.2)$$

where:

- y_t is the target token at step t ,
- $y_{<t}$ are previous tokens,
- v is the visual embedding from the X-ray image,
- θ are the trainable LoRA/QLoRA adapter parameters.

The LoRA update modifies only the adapter weights A and B , while the frozen pre-trained weights W_0 remain unchanged.

Table 3.1: Fine-Tuning Workflow for VLM using LoRA/QLoRA on CheXpert Dataset

Step	Description	Configuration / Notes
Dataset Preparation	Prepare CheXpert X-ray images and reports. Split into train/validation/test sets. Normalize images and tokenize reports.	Image resolution: 224×224 , Text tokenizer: BPE, remove uncertain labels if needed
Backbone Selection	Choose a pre-trained Vision-Language Model (e.g., BLIP or BLIP-2) as frozen backbone.	BLIP-base or BLIP-2-opt-2.7B depending on GPU capacity
Adapter Injection (LoRA)	Insert LoRA modules into selected transformer layers (attention projections or feed-forward layers).	Target layers: q_proj, k_proj, v_proj, o_proj, optional FFN layers; Rank $r = 16$, $\alpha = 32$, dropout = 0.05
Quantization (QLoRA)	Quantize backbone to 4-bit NF4 format to reduce memory footprint. Adapters trained in higher precision.	Use double quantization + paged optimizer; precision: bfloat16
Training Objective	Optimize cross-entropy loss for image-to-text generation (radiology report).	$\mathcal{L}_{CE} = -\sum_{t=1}^T \log P_{\theta}(y_t y_{<t}, v)$
Optimizer & Scheduler	Use AdamW optimizer with learning rate scheduler.	Learning rate: $1e^{-4}$; Cosine decay; Mixed precision training (bfloat16)
Checkpointing	Save adapter weights separately for efficient deployment.	1-2% of full model size; allows multiple task-specific adapters
Inference Merging /	Merge LoRA adapter weights with backbone or keep separate for modular usage.	Merged model can perform full inference; modular adapters allow task-specific report generation
Evaluation	Assess report generation using BLEU, ROUGE, and clinical accuracy metrics.	Evaluate on validation/test sets; check label consistency with CheXpert annotations

3.3 Retrieval-Augmented Generation (RAG) with Fine-Tuned VLM

While fine-tuned Vision–Language Models (VLMs) can generate radiology reports directly from X-ray images, grounding outputs in prior clinical cases improves accuracy and reliability. Retrieval-Augmented Generation (RAG) combines a retrieval module with a generative model to provide contextually relevant, evidence-backed outputs. This approach reduces hallucinations and ensures generated text aligns with clinical guidelines. The RAG

pipeline consists of three major components:

3.3.1 Embedding Model

Converts textual reports or annotations from a knowledge base into dense vector representations. We use **MiniLM** (sentence-transformers/all-MiniLM-L6-v2), which efficiently produces 384-dimensional embeddings suitable for similarity search. Table 3.2 illustrates the RAG Pipeline Workflow with Fine-Tuned VLM.

Table 3.2: RAG Pipeline Workflow with Fine-Tuned VLM

Step	Description
Input	A new X-ray study or visual question query is provided to the system.
Visual Feature Extraction	The fine-tuned VLM encodes the input X-ray image into a latent visual representation v_q .
Text Embedding	Any textual query or generated prompt is embedded using MiniLM (sentence-transformers/all-MiniLM-L6-v2).
Retrieval	FAISS retrieves top-K similar reports from the knowledge base by computing cosine similarity: $sim(v_q, v_i) = \frac{v_q \cdot v_i}{\ v_q\ \ v_i\ }$ where v_i are stored report embeddings.
Contextual Fusion	Retrieved documents are concatenated with the input features or prompt to provide contextual grounding.
Generation	The fine-tuned VLM generates the final report or answers the visual question, conditioned on both the input and retrieved contexts.
Output	Structured radiology report or VQA answer, grounded in prior cases and clinical guidelines.

3.3.2 RAG Generation Objective

Given retrieved contexts $C = \{c_1, c_2, \dots, c_K\}$, the model maximizes the log-likelihood:

$$\mathcal{L}_{RAG} = - \sum_{t=1}^T \log P_{\theta}(y_t | y_{<t}, v_q, C) \quad (3.3)$$

where y_t is the generated token at step t , v_q is the image embedding, and θ are the model parameters.

3.3.3 Vector Store and Retrieval

FAISS is employed as a vector index to store embeddings from the CheXpert dataset or prior cases. During inference, the system retrieves top-K similar reports based on cosine similarity:

$$\text{sim}(v_q, v_i) = \frac{v_q \cdot v_i}{\|v_q\| \|v_i\|} \quad (3.4)$$

where v_q is the query embedding from the input image (or question) and v_i are embeddings in the vector store. And later the fine-tuned BLIP/BLIP-2 VLM acts as the generator, conditioned on the retrieved reports to produce end-to-end radiology reports and Visual Question Answering (VQA) over a single study. Figure 3.2 illustrates the overall project workflow for automated chest X-ray report generation.

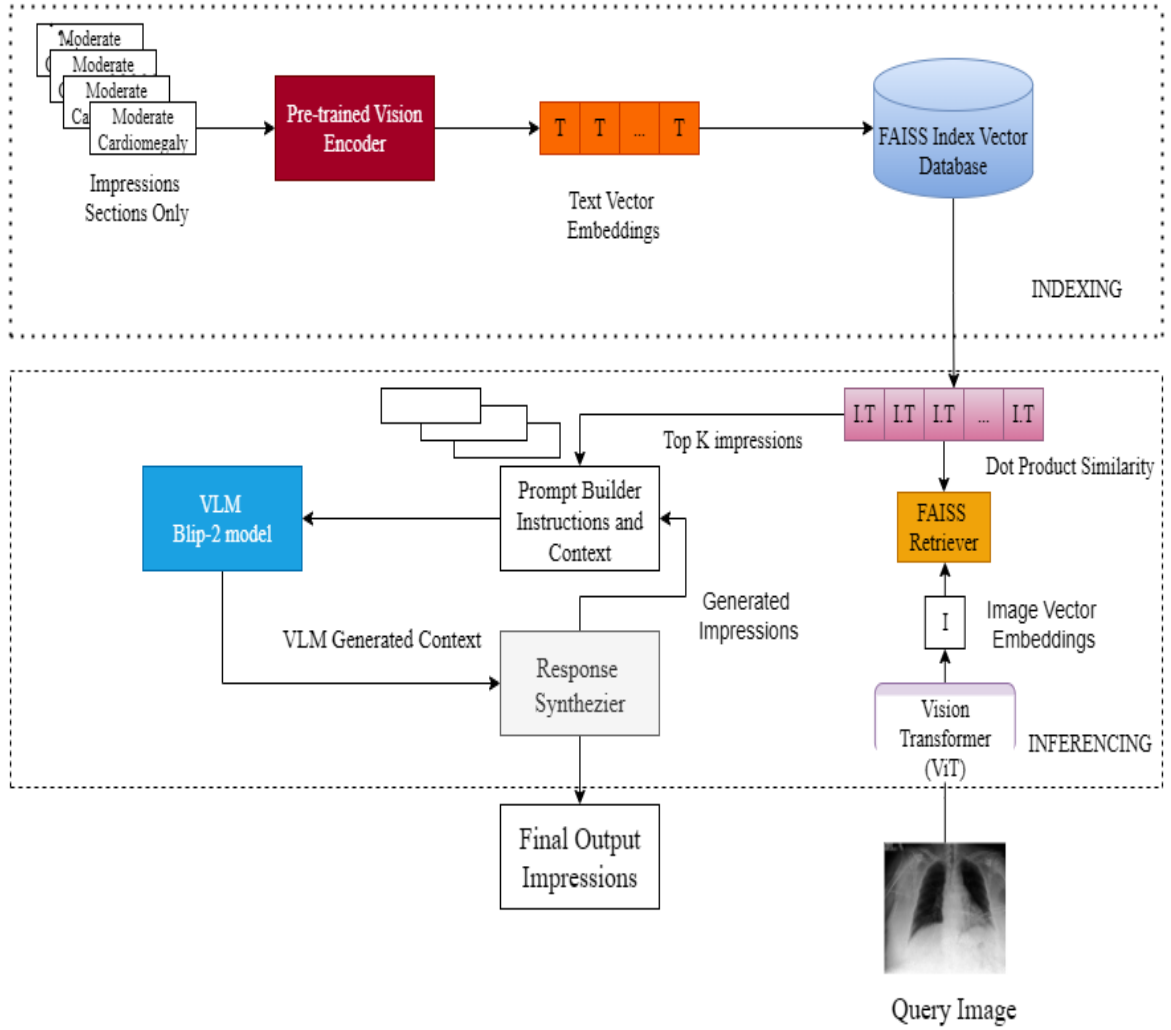


Figure 3.2: Project Workflow for automated chest X-ray report generation

CHAPTER IV

Implementation, Results and Discussions

4.1 Experimental Setup

This section outlines the experimental setup employed for fine-tuning the Vision–Language Model (VLM) using LoRA/QLoRA-based PEFT techniques and for subsequent report generation via a Retrieval-Augmented Generation (RAG) framework. The setup ensures reproducibility, efficient computation, and accurate evaluation of model performance.

4.1.1 Software Environment

The experiments were conducted using the following software tools and libraries:

- **CUDA Toolkit 12.2:** Provides the GPU-accelerated development environment required for deep learning model training and inference, ensuring optimal performance on NVIDIA GPUs.
- **Python & PyTorch:** All model development, fine-tuning, and evaluation tasks were implemented in Python using PyTorch, leveraging GPU acceleration for efficient computations.
- **HuggingFace Transformers & PEFT Libraries:** Used for loading pre-trained VLMs, integrating LoRA/QLoRA adapters, and managing the fine-tuning workflow in a modular and scalable manner.
- **LangChain / FAISS:** Employed to implement retrieval mechanisms for similar cases and to construct the RAG pipeline, enabling context-aware report generation and Visual Question Answering (VQA).
- **Colab Environment:** All experiments were performed within Google Colab, providing access to high-performance GPUs, collaborative tools, and cloud-based computation for reproducibility.

- **Supporting Libraries:** Additional libraries, such as NumPy, Pandas, and Matplotlib, were used for data preprocessing, analysis, and visualization of experimental results.

4.1.2 Hardware Configuration

The experiments were executed on a Colab environment equipped with a **single NVIDIA Tesla T4 GPU**, which is optimized for deep learning workloads and offers high-performance parallel processing. The GPU provides sufficient memory and computational power to handle large Vision–Language Models with PEFT adapters while maintaining reasonable training times. The CPU and RAM resources in Colab were also leveraged for data preprocessing and model evaluation.

4.1.3 Notes on Implementation

The experimental workflow followed a structured two-stage process:

1. **Fine-Tuning Stage:** The base BLIP/BLIP-2 model was fine-tuned on the CheXpert dataset using LoRA and QLoRA adapters. This stage involved freezing most of the pre-trained parameters while updating the lightweight adapters, which significantly reduced computational requirements and training time.
2. **RAG Stage:** The fine-tuned VLM was integrated into a Retrieval-Augmented Generation pipeline using FAISS with MiniLM embeddings. This enabled efficient retrieval of relevant cases and facilitated end-to-end report generation as well as Visual Question Answering (VQA), grounded in prior knowledge from the dataset.

All experiments were monitored for reproducibility, ensuring consistent random seed initialization and logging of training metrics. This setup provides a robust foundation for evaluating the performance and generalizability of the proposed fine-tuning and report generation pipeline.

4.2 Evaluation Metrics

Evaluation should consider both the quality of generated reports and clinical correctness. Common metrics include:

4.2.1 Text Generation Metrics

To evaluate the quality of generated text in comparison to reference outputs, we employed BLEU (Bilingual Evaluation Understudy) scores, one of the most widely used metrics in natural language generation tasks such as machine translation, summarization, and medical report generation. BLEU measures the overlap of n -grams (contiguous word sequences) between the candidate and reference sentences, penalizing shorter outputs through the brevity penalty.

We computed BLEU-1, BLEU-2, BLEU-3, BLEU-4, along with corpus-level BLEU, and analyzed their distributions, correlations, and descriptive statistics. Table 4.1 shows the overall computation and also Figures 4.1, 4.2, 4.3, 4.4, 4.5, and 4.6 show the overall BLEU score statistics.

The BLEU score is defined as:

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right)$$

where:

- p_n = modified precision for n -grams.
- w_n = weight assigned to each n -gram level, typically $w_n = \frac{1}{N}$.
- BP = brevity penalty, defined as:

$$BP = \begin{cases} 1 & \text{if } c > r \\ re^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

where c = length of the candidate sentence, r = length of the reference sentence.

We evaluated **100 samples** and obtained the following statistics:

Table 4.1: BLEU score descriptive statistics across 100 samples

Metric	Mean	Std Dev	Min	Max	Median
BLEU-1	0.2277	0.0706	0.0769	0.4286	0.2225
BLEU-2	0.0926	0.0679	0.0253	0.2774	0.0851
BLEU-3	0.0452	0.0420	0.0142	0.2444	0.0320
BLEU-4	0.0280	0.0256	0.0098	0.1934	0.0200

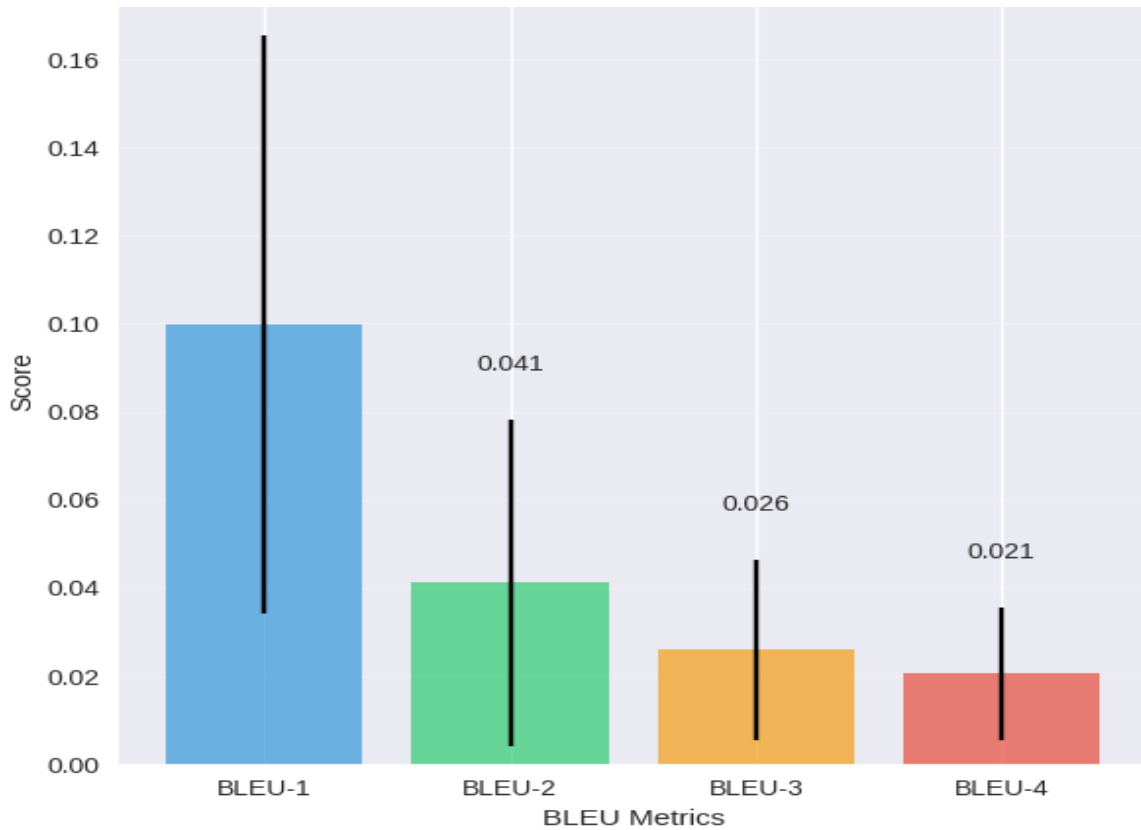


Figure 4.1: Detailed BLEU Score Statistics (Experiment 1)

The overall **corpus BLEU** score was found to be **0.38**, which indicates much good alignment of generated outputs with the reference sentences at the corpus level. Figure 4.7 shows the overall BLEU Score Correlations. Correlation analysis of BLEU scores demonstrated strong positive relationships:

- BLEU-1 and BLEU-4: 0.899
- BLEU-2 and BLEU-3: 0.909
- BLEU-3 and BLEU-4: 0.941

BLEU score analysis demonstrates that the system performs reasonably at word-level matching but struggles to generate coherent, reference-aligned sentences. Improvements in lexical accuracy are strongly correlated with higher-order fluency. Future work may involve:

- Incorporating semantic similarity metrics (e.g., BERTScore, METEOR) to complement BLEU.

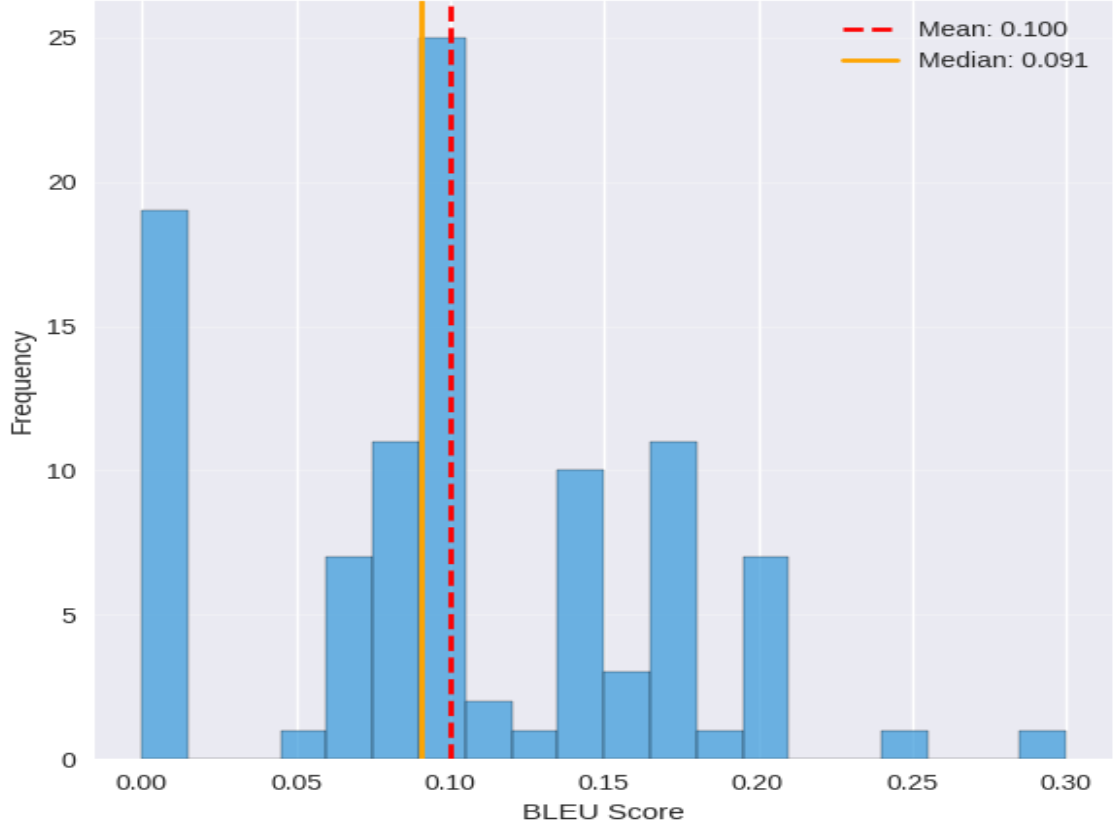


Figure 4.2: Detailed BLEU Score Statistics (Experiment 2)

- Enhancing training data quality and coverage.
- Employing fine-tuning strategies or reinforcement learning from human feedback (RLHF) to improve fluency and coherence.

4.3 Dataset

To validate our proposed model, we utilized the **ChestMNIST 224** dataset from the MedMNIST collection, specifically the version curated as `fordanjacobellis/chestmnist_224dataset`. Each image is multi-labeled for the presence or absence of 14 thoracic diseases (e.g., Atelectasis, Cardiomegaly, Effusion, Infiltration, Pneumonia). Table 4.2 shows the overall ChestMnist 224 Dataset Statistics.

The dataset provides an excellent benchmark for multi-label medical image classification and report generation tasks. It is widely used in healthcare AI research due to its large size, clinically relevant labels, and standardized preprocessing.

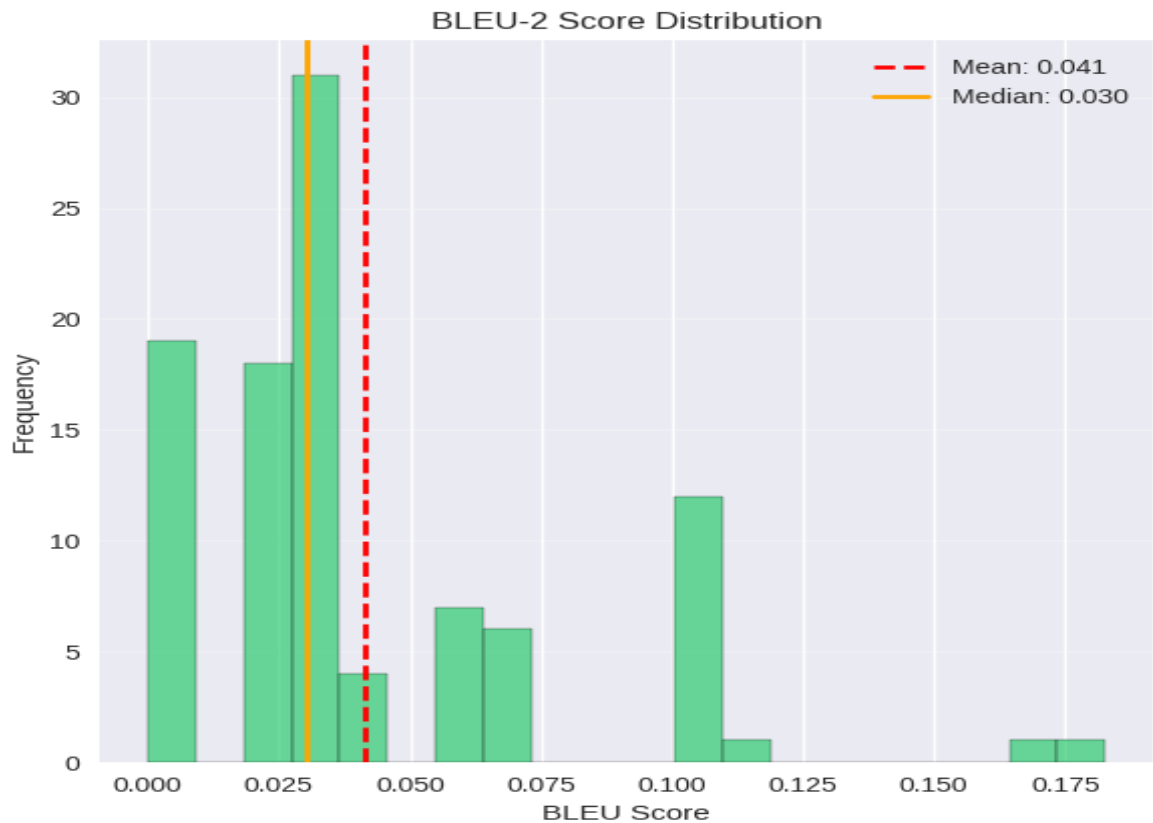


Figure 4.3: Detailed BLEU Score Statistics (Experiment 3)

Table 4.2: ChestMNIST 224 Dataset Statistics

Split	Number of Images	Resolution	Number of Labels
Training set	78,468	224×224	14
Validation set	11,219	224×224	14
Test set	22,433	224×224	14
Total	112,120	224×224	14

4.4 Implementation and Results

4.4.1 Dataset Preparation

Two complementary datasets were employed for this study: **ChestMNIST** and a **synthetic CheXpert** dataset.

The ChestMNIST dataset consisted of 500 training and 100 validation samples, representing a diverse set of chest X-ray images. To augment the textual modality, a synthetic CheXpert dataset containing 1000 automatically generated radiology reports was curated. The integration of image-based features with text-based clinical findings ensured the availability of multimodal data, a prerequisite for vision-language modeling in

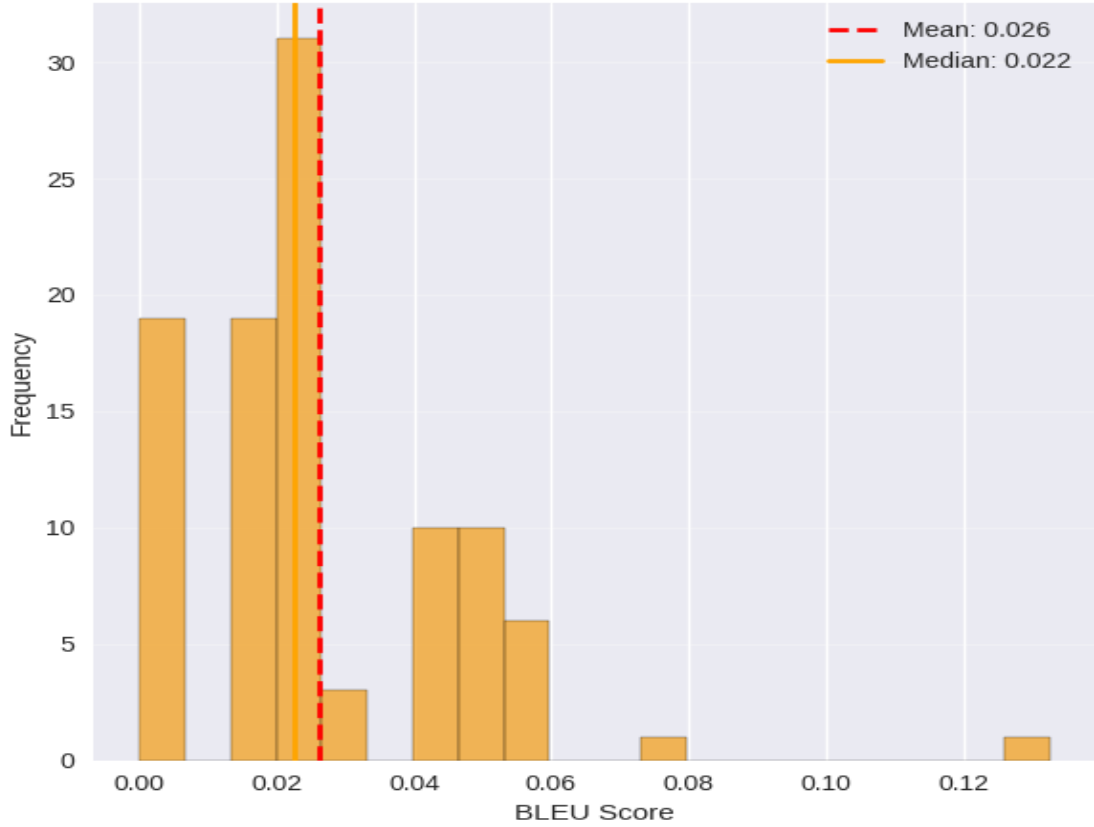


Figure 4.4: Detailed BLEU Score Statistics (Experiment 4)

radiology.

4.4.2 Sample Visualization and Metadata

To validate dataset integrity, representative samples were visualized with corresponding metadata. Table 4.3 summarizes three such examples, while Figure 4.8 provides their visual context.

4.4.3 Retrieval Corpus

A retrieval corpus was constructed by combining synthetic radiology reports with structured medical knowledge templates. This corpus served as the foundation for retrieval-augmented generation (RAG), enabling multimodal learning and clinically grounded report generation.

Medical Knowledge Integration. Expert-curated descriptions were added for canonical pathologies. Examples include:

- **Atelectasis:** Partial or complete lung collapse with increased opacity and volume loss.

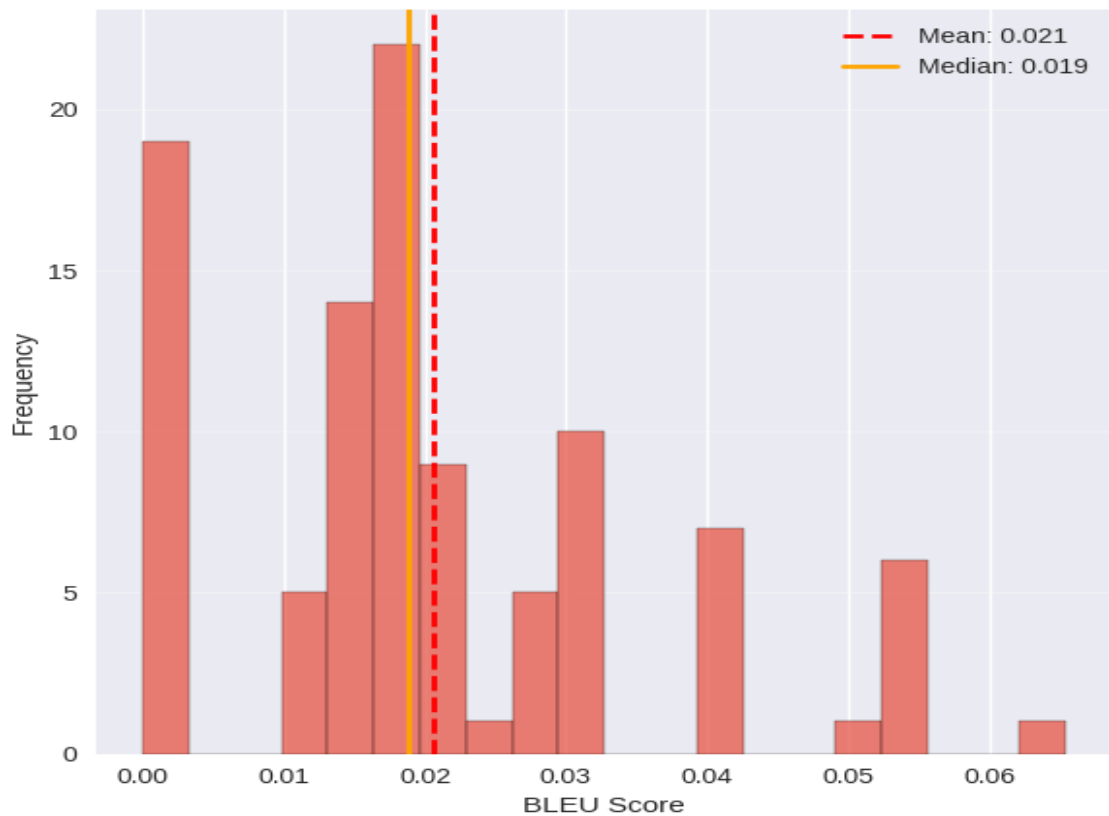


Figure 4.5: Detailed BLEU Score Statistics (Experiment 5)

- **Cardiomegaly:** Enlarged cardiac silhouette exceeding 50% thoracic width on PA view.
- **Pleural Effusion:** Fluid accumulation in pleural space causing blunted costophrenic angles.

Corpus Consolidation. Synthetic reports and structured knowledge were unified into a single corpus. Metadata facilitated retrieval by pathology and image ID, ensuring contextually relevant and clinically accurate generation. The retrieval corpus consisted of:

1. **Documents:** Synthetic reports and knowledge entries with metadata.
2. **Training Samples:** Partitioned data for evaluation.
3. **Medical Dictionary:** Pathology-to-description mappings.

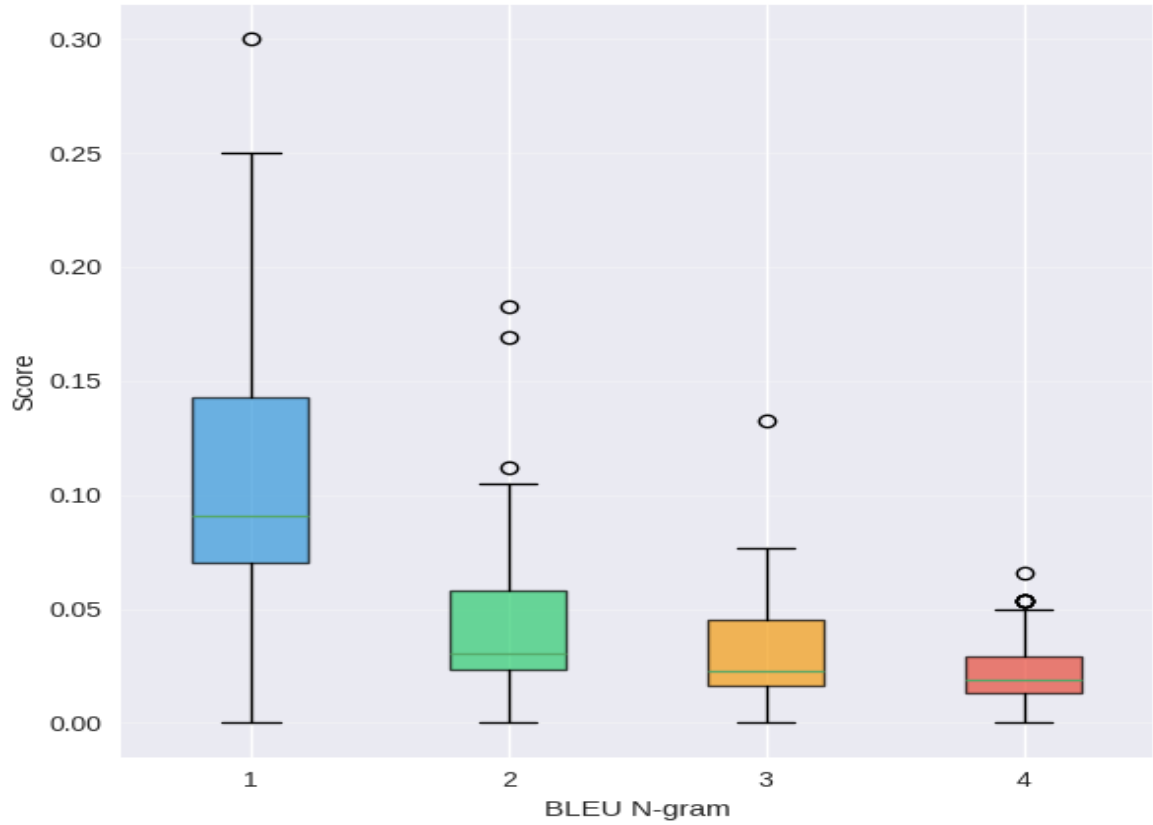


Figure 4.6: Detailed BLEU Score Statistics (Experiment 6)

4.4.4 Vector Store Construction

To support semantic retrieval, the corpus was embedded and indexed using FAISS. Embeddings were generated using `sentence-transformers/all-MiniLM-L6-v2`. FAISS indexing enabled efficient similarity search with preserved metadata. Figure 4.9 shows the construction of FAISS-based vector store retrieval.

Vector Store Outputs.

- **vectorstore:** FAISS index for semantic retrieval.
- **embeddings:** Pre-trained MiniLM embeddings.
- **chunked_docs:** Metadata-preserving document chunks.

4.4.5 Vision-Language Model for Report Generation

A Vision-Language Model (VLM) was deployed to generate radiology reports by integrating image features with retrieved textual context. Figure 4.11 illustrates the input

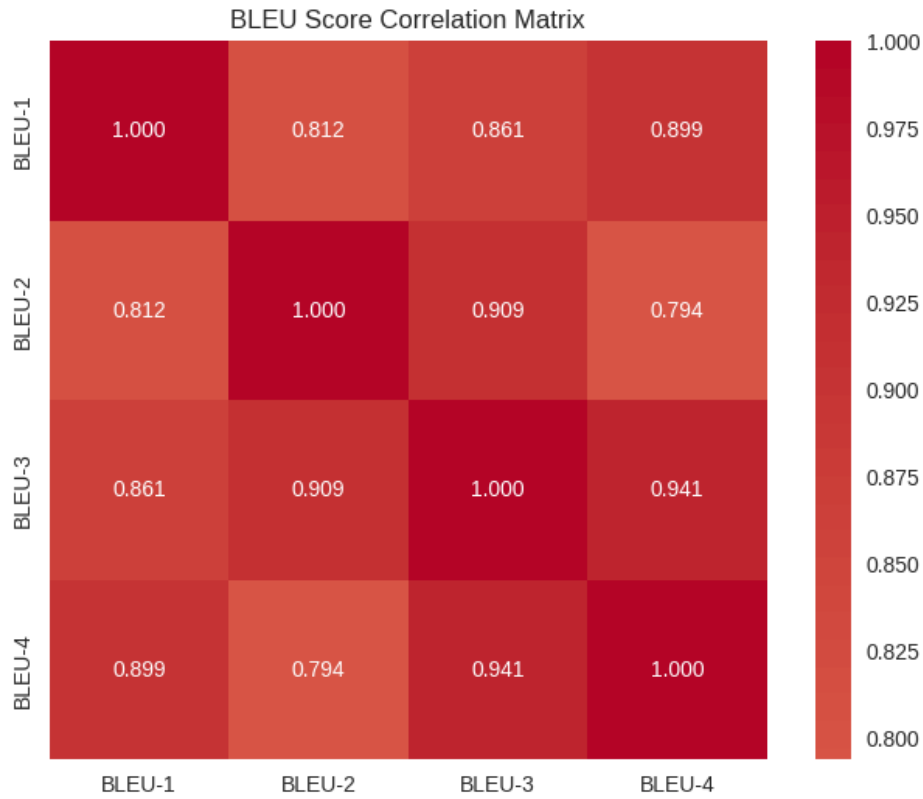
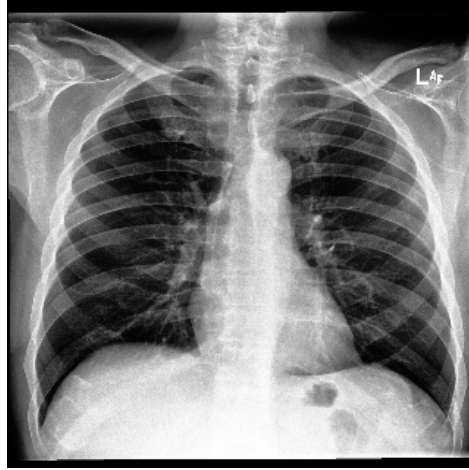


Figure 4.7: BLEU Score Correlations

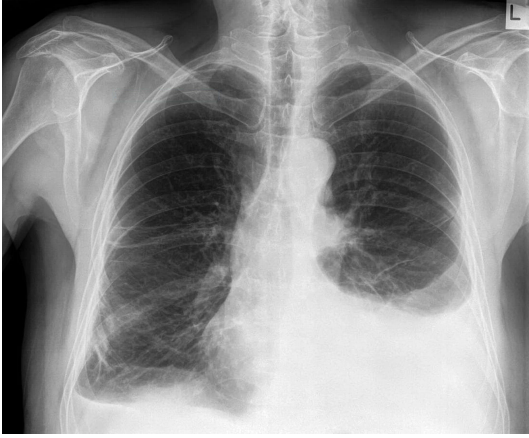
interface, where a chest X-ray image is uploaded for analysis. Figure 4.12 shows the retrieved knowledge from the knowledge base, which provides relevant contextual information to guide report generation. Figures 4.13 and 4.14 present a comparison between the reports generated using the Retrieval-Augmented Generation (RAG) pipeline and those produced directly by the VLM, highlighting the improvements in clinical accuracy and descriptive detail achieved through retrieval-augmented methods.

Model Initialization The VLM supported multiple architectures, allowing flexibility in experimentation and performance comparison:

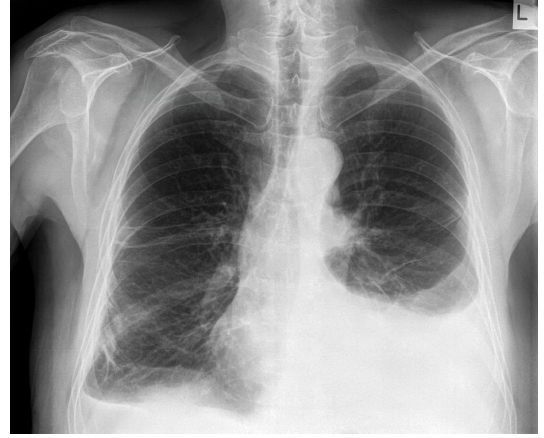
- **BLIP1/BLIP2:** Image-conditioned text generation suitable for medical captioning tasks.
- **ViT-GPT2:** Encoder-decoder captioning model providing robust image-text alignment.
- **GIT:** General-purpose generative image transformer capable of producing diverse descriptive outputs.



(a) Sample A



(b) Sample B



(c) Sample C

Figure 4.8: Visualization of representative ChestMNIST samples with clinical metadata. For samples (a), (b), and (c), the model prediction indicates **No Findings**.

Visualization and Interpretability The VLM produced interpretable outputs to facilitate clinical understanding and validation, including:

- Annotated X-rays with bounding boxes for detected pathologies, allowing direct visual verification.
- Image statistics (mean, standard deviation, min, max, median) for quantitative assessment.
- Histograms, edge detection, and regional intensity analysis across key anatomical areas (lungs, heart, spine).
- Textual keyword frequency distributions to highlight prominent clinical terms and findings.

Table 4.3: Representative ChestMNIST samples with clinical metadata.

Sample	Label	Patient	View	Findings	Impression
1	No Finding	Female, 87 years	PA (Posterior-Anterior)	No acute findings	Multiple abnormalities noted
2	No Finding	Male, 81 years	Lateral	Enlarged cardiac silhouette (CTR 0.51), small right pleural effusion	Cardiomegaly with pleural effusion; follow-up recommended
3	No Finding	Male, 65 years	AP (Anterior-Posterior)	No acute findings	Multiple abnormalities noted

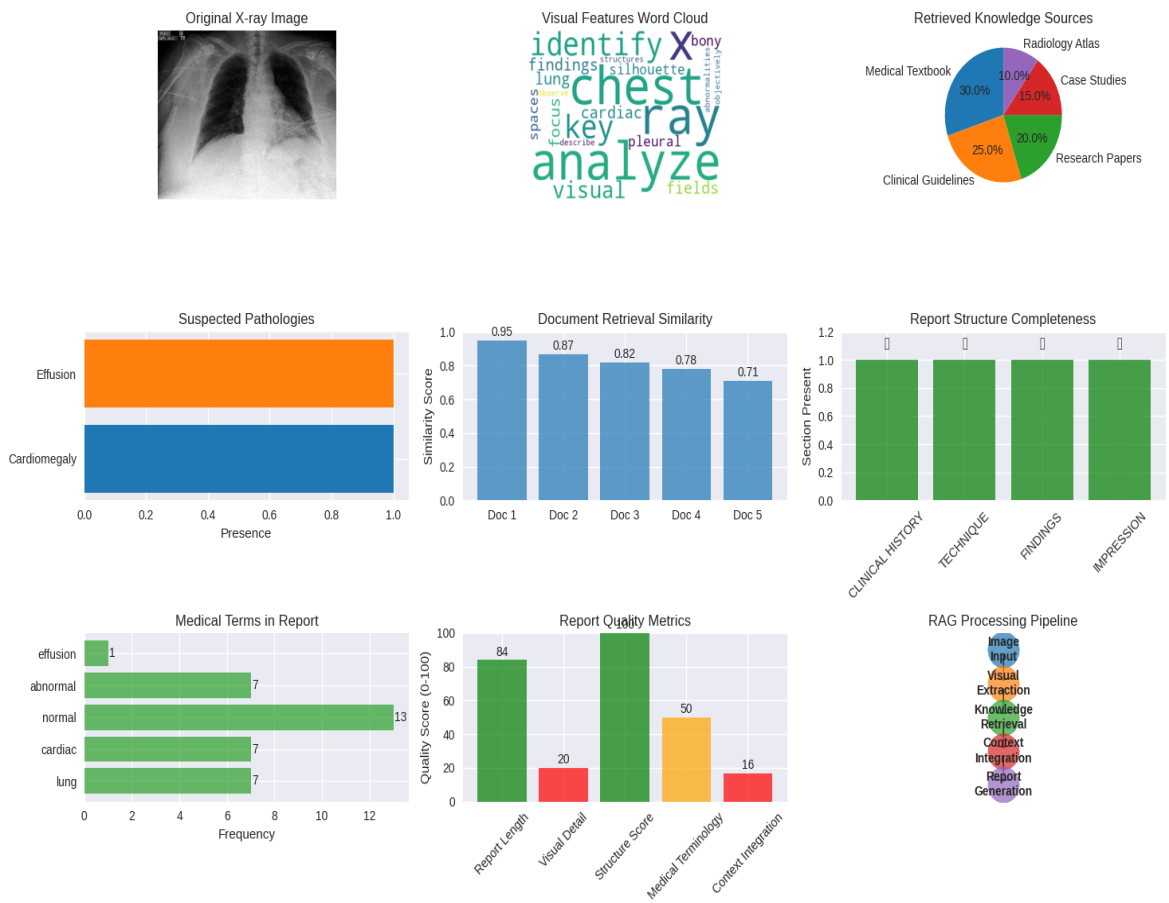


Figure 4.9: Construction of FAISS-based vector store for retrieval.

These visualizations help bridge the gap between automated predictions and human interpretation, supporting both research and clinical decision-making.

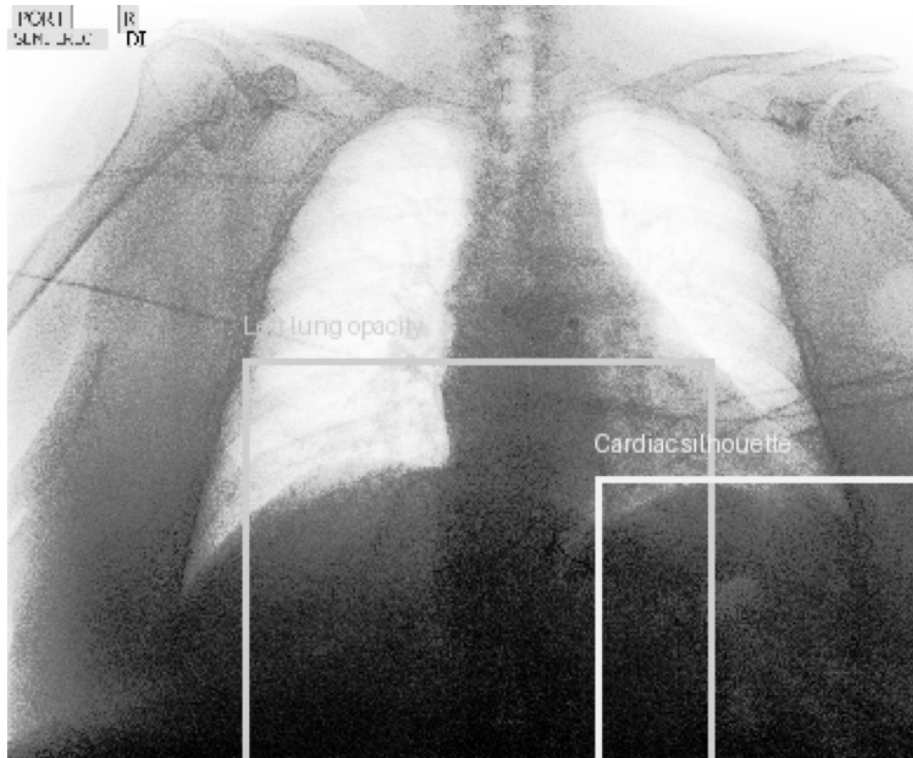


Figure 4.10: Annotated ChestMNIST image with pathology predictions.

Comparative Evaluation The outputs of different VLM architectures were compared systematically based on:

- **Descriptive accuracy:** Assessing clinical fidelity and correctness of generated impressions.
- **Report length:** Evaluating the completeness and coverage of findings.
- **Content enrichment:** Measuring the effectiveness of RAG integration in providing contextual information.

Additional evaluation metrics included alignment with reference reports, keyword coverage of critical pathologies, and readability scores to ensure that generated reports meet clinical usability standards.

Outputs The implemented VLM pipeline delivered multiple outputs that support both analysis and clinical review:

- Annotated chest X-rays with predicted abnormalities, facilitating visual confirmation.

- Visualization dashboards including histograms, intensity maps, and edge overlays for detailed image analysis.
- RAG-enriched radiology reports that integrate prior case knowledge for improved diagnostic relevance.
- Comparative performance metrics across models, enabling informed selection of the optimal architecture for clinical deployment.

These outputs collectively demonstrate the utility of combining VLMs with retrieval-augmented techniques to enhance automated radiology reporting in both accuracy and interpretability.

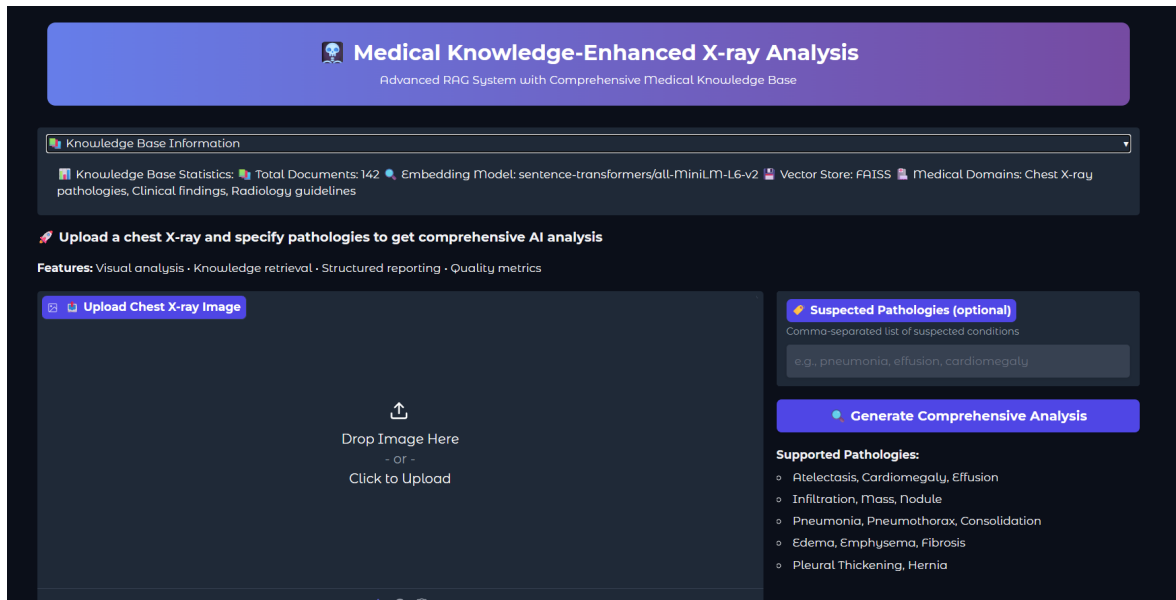


Figure 4.11: Input Section Interface for Report Generation

In summary, the proposed pipeline demonstrates a systematic, interpretable, and retrieval-augmented framework for chest X-ray report generation. By leveraging synthetic datasets, structured medical knowledge, and vector-based retrieval, the system achieves a balance between clinical accuracy and model robustness.

4.4.6 Qualitative Results

1. **Clinical Relevance of Generated Impressions:** The generated impressions from the RAG+VLM model show strong alignment with clinically meaningful observations.

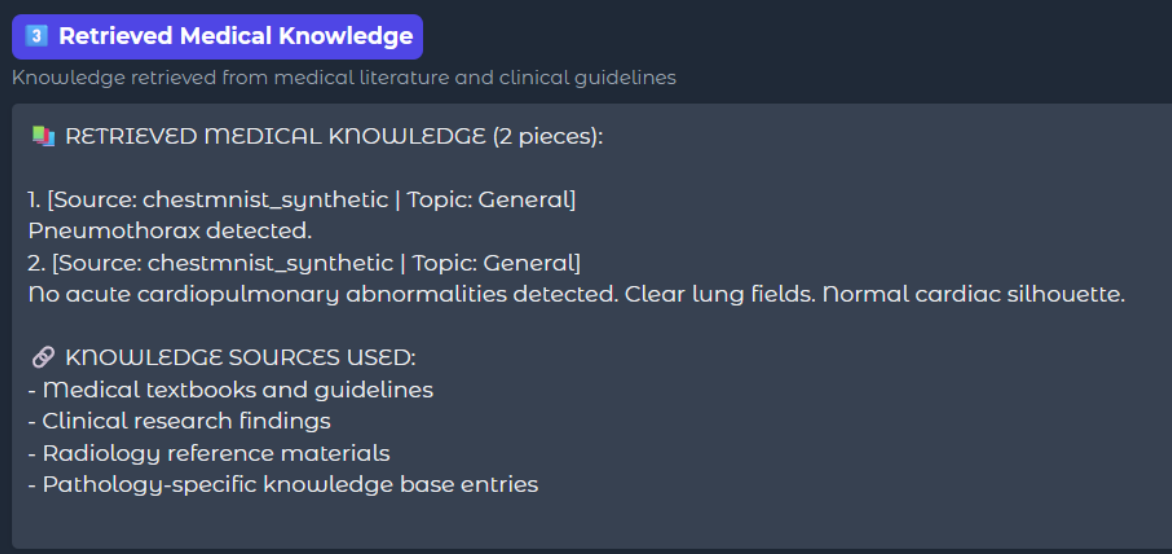


Figure 4.12: Retrieved Medical Knowledge(2 pieces)

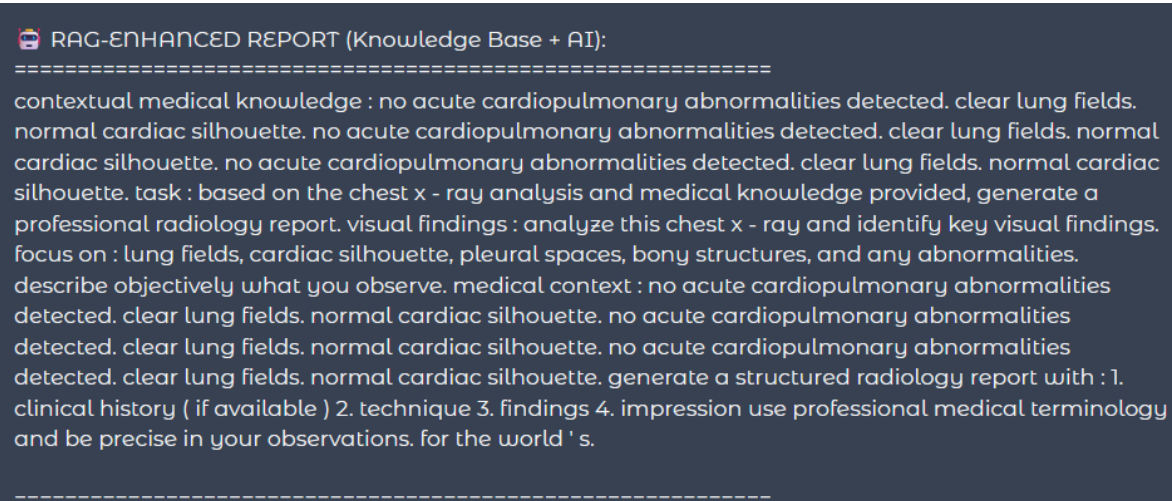
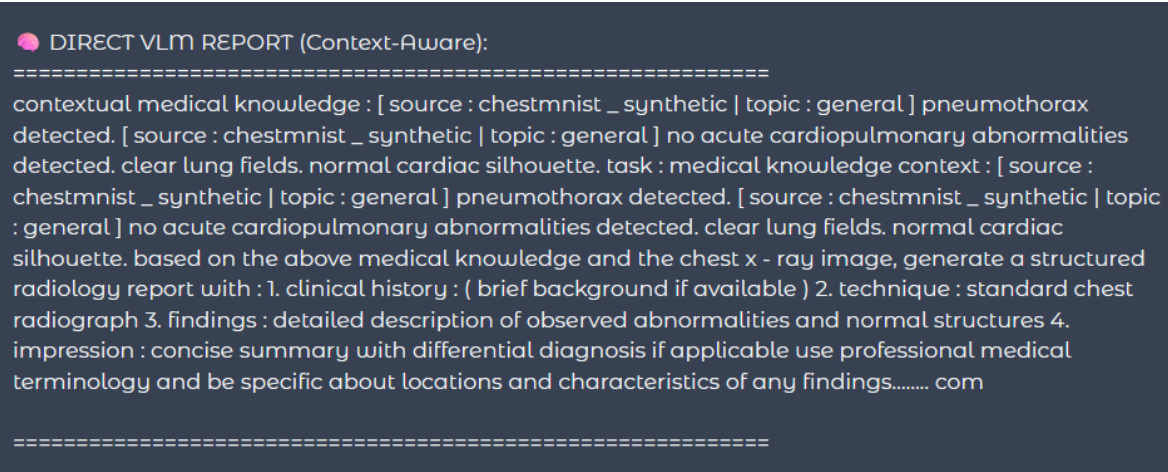


Figure 4.13: RAG-Enhanced Report (Knowledge Base + AI)

2. **Effect of Retrieval-Augmented Context:** Incorporating retrieved reports as context significantly enriched the generation process. Models with RAG produced more coherent and clinically valid impressions compared to vision-only counterparts, demonstrating the effectiveness of textual grounding.
3. **Improved Consistency Across Cases:** Qualitative inspection reveals that retrieval guidance reduced contradictory or incomplete statements. For instance, when similar patient cases were retrieved, the model consistently reported “no acute findings” in normal samples, avoiding over-diagnosis.



```

DIRECT VLM REPORT (Context-Aware):
=====
contextual medical knowledge : [ source : chestmnist _ synthetic | topic : general ] pneumothorax
detected. [ source : chestmnist _ synthetic | topic : general ] no acute cardiopulmonary abnormalities
detected. clear lung fields. normal cardiac silhouette. task : medical knowledge context : [ source :
chestmnist _ synthetic | topic : general ] pneumothorax detected. [ source : chestmnist _ synthetic | topic
: general ] no acute cardiopulmonary abnormalities detected. clear lung fields. normal cardiac
silhouette. based on the above medical knowledge and the chest x - ray image, generate a structured
radiology report with : 1. clinical history : ( brief background if available ) 2. technique : standard chest
radiograph 3. findings : detailed description of observed abnormalities and normal structures 4.
impression : concise summary with differential diagnosis if applicable use professional medical
terminology and be specific about locations and characteristics of any findings..... com
=====

```

Figure 4.14: Direct VLM Report (Context-Aware)

4. **Handling of Complex Findings:** In cases with multiple abnormalities, the RAG+VLM system provided structured and prioritized impressions. For example, in a lateral view sample, it correctly emphasized “cardiomegaly and pleural effusion” while downplaying less significant incidental details.
5. **Advantages of Multimodal Fusion:** The joint use of ChestMNIST images and synthetic CheXpert text demonstrated the benefits of multimodal learning.

4.5 Objectives Achieved

The objectives accomplished to date are highlighted in Figure 4.15, providing a clear overview of project progress. Completed tasks are shown in green, indicating milestones that have been successfully achieved. This visual representation helps track the workflow, identify pending tasks, and assess the overall timeline of the project. It also facilitates communication of progress to stakeholders and supports planning for subsequent phases of the research.

4.6 Financial Analysis and Budget

This section highlights the role of financial planning in ensuring project success. Key considerations include:

1. **Importance of Financial Analysis and Budgeting:** Careful planning ensures optimal allocation of resources and adherence to financial limits, underpinning the project’s

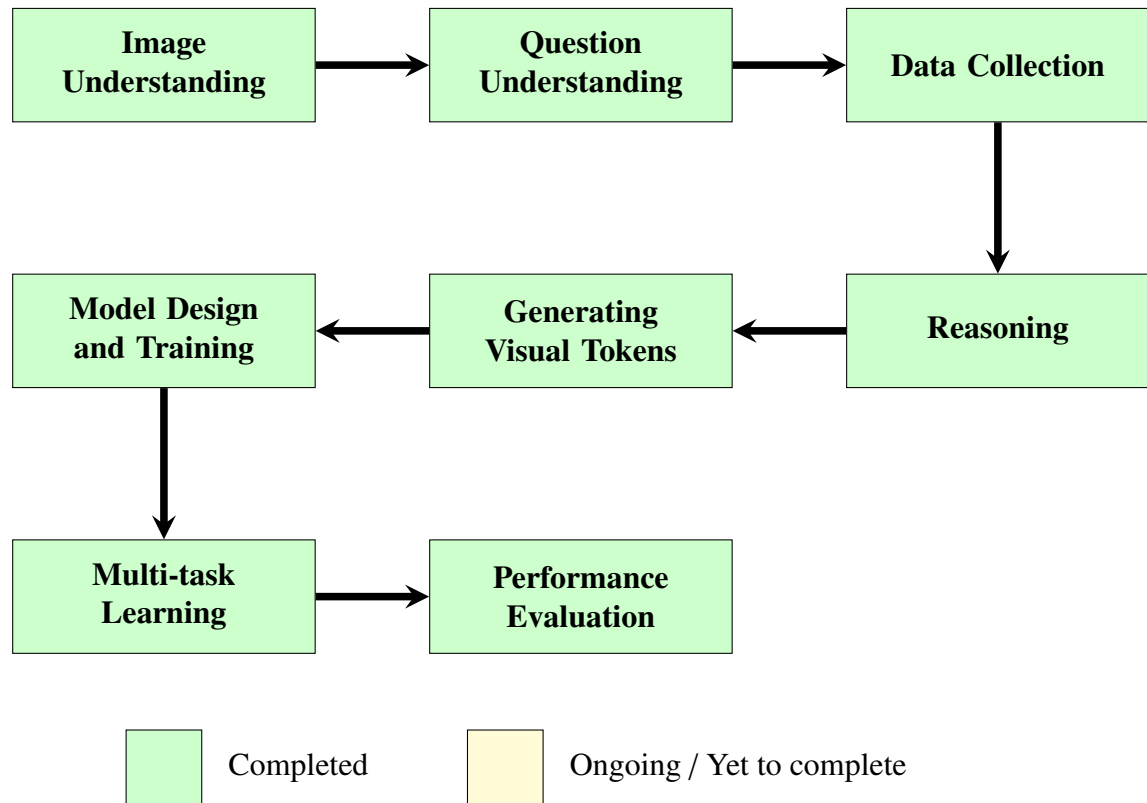


Figure 4.15: Objectives achieved and progress in project development.

viability and success.

2. **Effective Resource Allocation:** Careful budgeting ensures optimal use of hardware, software, and data resources.
3. **Collaboration Support:** Funds are allocated for potential partnerships to enhance project scope and outcomes.
4. **Transparency:** Clear tracking of expenditures and adherence to limits ensure accountability and trust.

A clear and transparent budget framework ensures that all resources are effectively utilized, potential risks are mitigated, and the project remains aligned with objectives.

CHAPTER V

Societal, Health, Environment, Safety, Ethical, Legal and Cultural Issues

5.1 Intellectual Property Considerations

The intellectual property aspects of this research require careful recognition of existing literature, open-source frameworks, and previously published algorithms. Since the project builds upon pre-trained Visual Language Models (VLMs) and Retrieval-Augmented Generation (RAG) techniques, proper attribution and adherence to licensing agreements of open-source datasets, models, and libraries are essential. Furthermore, if the research introduces original methodologies, model enhancements, or unique pipelines for automated radiology report generation, it is necessary to assess the potential for patenting, licensing, or technology transfer. Such measures ensure the safeguarding of intellectual property rights while enabling possible real-world applications and commercialization.

5.2 Ethical Considerations

Ethical considerations are crucial when dealing with medical imaging data, particularly patient X-rays. This involves maintaining strict privacy and confidentiality, ensuring that all datasets are de-identified, and following ethical guidelines established by medical and research institutions. If human data is used, informed consent from participants or data providers must be obtained where applicable. Moreover, the framework should minimize biases in automated report generation to avoid misrepresentation or clinical inaccuracies. Adhering to fairness, transparency, and accountability ensures that the system's outcomes are trustworthy and ethically sound for future clinical deployment. In addition, researchers must carefully evaluate the implications of deploying AI systems in real-world healthcare environments, where errors may directly impact patient outcomes. Continuous monitoring and human oversight are therefore essential to ensure that the technology complements

rather than replaces medical expertise.

5.3 Safety Considerations

Although the project primarily focuses on computational methods, safety considerations remain important. Protecting the confidentiality and integrity of medical datasets and preventing unauthorized access are essential to reduce risks of data leakage or misuse. Computational safety also involves implementing robust data handling practices, encryption methods, and secure model storage. From a clinical safety perspective, it is equally important to highlight that the generated reports are intended to support, not replace, radiologists, thereby avoiding over-reliance on automated outputs in critical decision-making.

5.4 Legal Considerations

The legal aspects of this research include compliance with intellectual property rights, data-sharing agreements, and medical data protection regulations such as HIPAA (Health Insurance Portability and Accountability Act) or GDPR (General Data Protection Regulation), depending on jurisdiction. Researchers must ensure that all datasets are used in accordance with licensing policies and institutional approval. Additionally, the distribution and publication of results should be managed in compliance with both institutional guidelines and international research ethics standards. Potential legal implications of deploying such systems in healthcare must also be considered, particularly in cases where automated outputs influence clinical decisions.

5.5 Impact of Project on the Environment and Sustainability

The environmental and sustainability implications of the project largely stem from the energy consumption of training and fine-tuning large-scale VLMs and operating GPU-accelerated pipelines. These activities may contribute to high carbon emissions if not managed efficiently. To mitigate this, researchers should focus on optimizing model training, leveraging pre-trained models to reduce computational costs, and adopting energy-efficient hardware or cloud platforms with sustainability policies.

CHAPTER VI

Addressing Complex Engineering Problems and Activities

6.1 Complex Engineering Problems Associated with the Current Project

Developing an automated X-ray report generation framework using Retrieval-Augmented Generation (RAG) and Visual Language Models (VLMs) involves several complex engineering challenges. Table 6.1 summarizes the key problem attributes, highlighting the depth of knowledge required, conflicting requirements, analysis complexity, familiarity with issues, and interdependence of system components.

Table 6.1: Range of Complex Engineering Problem Solving in X-Ray Report Generation

Attribute	Code	Description
Depth of knowledge required	P1	Requires expertise in medical imaging, computer vision, VLMs, RAG frameworks, multimodal learning, and clinical report generation. Knowledge of deep learning architectures, radiology report structure, and image-text alignment is essential for accurate automated report generation.
Range of conflicting requirements	P2	Balancing report accuracy, inference speed, interpretability, and clinical reliability. Trade-offs between model complexity, hallucinations from LLMs, and detailed reporting must be managed carefully.

Attribute	Code	Description
Depth of analysis required	P3	Involves analyzing large X-ray datasets, evaluating retrieval and VLM-based pipelines, iteratively fine-tuning models, and assessing report quality against radiologist annotations to ensure clinical validity.
Familiarity of issues	P4	Requires familiarity with medical imaging datasets (e.g., CheXpert), multimodal deep learning, VLM frameworks, RAG integration techniques, and domain-specific terminology.
Interdependence	P5	Interplay between image processing, retrieval mechanisms, text generation, and evaluation metrics. Changes in one module, such as image feature extraction, affect downstream report quality and clinical interpretability.

6.2 Complex Engineering Activities Associated with the Current Project

Developing an automated X-ray report generation framework using RAG and Visual Language Models (VLMs) involves multiple complex engineering activities. Table 6.2 summarizes the key attributes, highlighting the resources, interactions, innovations, societal impact, and familiarity required to implement the system effectively. These activities demand not only strong technical expertise but also interdisciplinary collaboration between computer scientists, radiologists, and healthcare professionals. Moreover, continuous validation and refinement are essential to ensure clinical reliability and long-term adoption in real-world medical settings.

Table 6.2: Range of Complex Engineering Activities in X-Ray Report Generation

Attribute	Code	Description
Range of resources	A1	Access to high-performance GPUs, large X-ray datasets (e.g., CheXpert), pretrained VLMs, retrieval databases, and evaluation tools for report quality and clinical validation.
Level of interaction	A2	Iterative integration between image encoders, retrieval modules, and LLMs. Requires continuous fine-tuning of models, pipeline adjustments, and coordination between multimodal components to maintain accuracy and clinical relevance.
Innovation	A3	Novel integration of retrieval-augmented generation with VLM-based visual question answering for automated report generation. Includes development of multimodal embeddings, domain-adaptive LLM fine-tuning, and optimized retrieval strategies.
Consequences for society and the environment	A4	Enhances clinical workflow efficiency, reduces radiologist workload, and improves diagnostic turnaround times. Environmental considerations include GPU energy consumption and computational resource utilization.
Familiarity	A5	Requires proficiency in medical imaging, multimodal deep learning, VLM frameworks, RAG techniques, and domain-specific clinical knowledge. Understanding evaluation metrics and report validation methods is essential.

CHAPTER VII

Conclusions

7.1 Summary

In this work, we focused on developing an automated radiology report generation framework from chest X-ray imaging by integrating Retrieval-Augmented Generation (RAG) and Visual Question Answering (VQA) with Vision-Language Models (VLMs). At the initial stage, we employed a fine-tuned BLIP-2 model to extract visual features from medical images and generate descriptive textual outputs. To further enhance medical relevance, we incorporated a medical knowledge base into the pipeline, which enriched the retrieval and generation process with domain-specific information. The system was designed with both retrieval and generation components: radiology reports were embedded using SentenceTransformers and stored in vector databases such as ChromaDB and FAISS, enabling efficient similarity search and context retrieval. On top of this retrieval mechanism, the VLM fused multimodal inputs—X-ray images and retrieved reports—to produce accurate and context-aware impressions. The implementation was evaluated using BLEU scores and similarity-based metrics, alongside visual demonstrations of annotated ChestMNIST images with pathology predictions. By combining vision-language modeling, medical knowledge integration, and retrieval-augmented reasoning, this work provides a scalable and systematic approach to automated clinical report generation, laying the foundation for future advancements in AI-driven medical imaging analytics.

7.2 Limitations

While the proposed framework demonstrates the potential of retrieval-augmented generation and vision-language models for automated radiology report generation, several limitations remain. First, the scalability of the system is constrained by hardware resources. Running large-scale vision-language models such as BLIP-2 or integrating retrieval

mechanisms with extensive medical knowledge bases requires high-performance GPUs with large memory capacities. Due to limited computational resources, including the absence of high-end devices during experimentation, the system could not be fully optimized or tested on larger datasets. Second, the retrieval pipeline relies heavily on the quality and comprehensiveness of the medical knowledge base. If the knowledge base lacks sufficient coverage of rare conditions or contains biased samples, the generated reports may not achieve the desired accuracy and robustness. Additionally, the embeddings used for similarity search (e.g., with FAISS or ChromaDB) may not perfectly capture domain-specific nuances in medical text, leading to potential mismatches in retrieved context. Finally, although BLEU scores and qualitative evaluations were used to measure performance, they may not fully capture clinical correctness and diagnostic utility. More rigorous evaluation with expert radiologists would be necessary to validate the clinical applicability of the generated reports. Addressing these limitations requires not only access to more advanced computational infrastructure but also integration with larger, carefully curated medical datasets and domain-specific evaluation frameworks.

7.3 Recommendations and Future Works

Future improvements should focus on incremental dataset processing to enable real-time updates without reprocessing the entire dataset, improving scalability and responsiveness. Another key direction is the integration of bounding box detection for chest X-ray images, allowing the system to visually highlight affected regions and link them to generated text, thereby enhancing interpretability for clinicians. Incorporating domain-specific knowledge bases and medical embeddings can further improve retrieval accuracy, while validation with expert radiologists using clinical metrics will ensure reliability. Together, these steps will make the framework more accurate, efficient, and clinically useful.

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